

Olusegun Odeyemi

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Executive Summary

Security Engineer & Technical Founder driving the convergence of machine learning, decentralised finance (DeFi), and enterprise risk architecture. With a proven track record of architecting deep-tech solutions from theoretical mathematical models to production-ready software, I specialise in building high-throughput compliance engines capable of processing millions of concurrent transactions with sub-second latency.

As the creator and lead architect of the **Anti-Money Laundering Transaction Trust Protocol (AMTTP)**, I engineered a full-stack, hybrid infrastructure (Next.js, Flutter, FastAPI, Solidity) that leverages distilled machine learning student models (XGBoost) and Zero-Knowledge Proofs (ZK-SNARKs) to solve systemic regulatory blindness in Web3. Alongside that production platform, I developed my proprietary mathematical framework, **Blind-Spot Decomposition Theory (BSDT)**, as a separate research and optional enterprise analytics capability for modelling hidden failure modes and distributed system instability.

Recognised by industry leaders for bridging complex algorithmic research with commercial application, I bring hands-on expertise in secure-by-design systems, scalable API gateways, and multi-domain threat intelligence. I am uniquely positioned to advance the UK's global leadership in secure digital infrastructure, regulatory technology, and applied artificial intelligence.

Core Competencies

Security Engineering & Technical Architecture

- Secure-by-design application development, API gateway security, and zero-knowledge identity protocols
- Hardening cloud infrastructure and containerised deployments across distributed microservices
- Penetration testing (black-box), vulnerability exploitation, and programmatic remediation of OWASP Top 10 flaws

Machine Learning & Automated Fraud Detection

- Engineering high-accuracy predictive models (XGBoost, Graph ML) for real-time anomaly detection
- Processing and classifying millions of concurrent data payloads with high inference speeds
- Architecting automated regulatory policy engines that adapt to high-velocity transactional environments

Data Systems & Secure Enterprise Integrations

- Managing secure multi-million revenue data flows and third-party API dependencies
- Implementing robust Role-Based Access Control (RBAC) across multiple frontend clients
- Engineering custom threat intelligence dashboards and active transaction monitoring systems

Professional Experience

Founder & Lead Architect (AMTTP)

August 2023 – Present

Derby, United Kingdom

- Engineered and deployed a full-stack decentralised compliance MVP integrating Python microservices, a Next.js administrative dashboard, a Flutter consumer client, and Solidity smart contracts.

- Built a real-time ML inference pipeline evaluated on 625,168 Ethereum address samples, deploying CPU-optimised student models with ROC-AUC 0.9898 and near-perfect ensemble validation under knowledge distillation.
- Published four research papers across TechRxiv and Zenodo, including BSDT, Universal Detection Principle (UDP), and the Geometry of System Collapse, establishing a coherent research programme spanning fraud detection, anomaly representation, and systemic instability.
- Filed a UK patent protecting the core cryptographic and machine learning methodology underlying distributed transaction compliance.

Technical Account Manager (Logistics & Enterprise APIs) *October 2022 – February 2023*
Jumia, Lagos, Nigeria

- Managed strategic technical relationships for top-tier enterprise vendors, architecting complex supply chain API integrations that streamlined high-throughput inventory routing and drove millions of dollars in sustained e-commerce revenue growth.
- Bridged the gap between commercial partners and internal engineering teams by automating real-time data flow pipelines, significantly reducing delivery latency and optimizing last-mile logistics across multi-hub fulfillment ecosystems.
- Protected multi-million revenue payment streams by auditing client integration protocols, identifying deep architectural vulnerabilities, and enforcing stringent API operational controls for commercial accounts.

Cyber Security Engineer & Integration Specialist *October 2017 – July 2021*
Mctoolu Global Concept, Lagos, Nigeria

- Programmed and executed automated security testing scripts and manual penetration tests (black-box) against critical third-party web application integrations.
- Exploited and subsequently remediated critical OWASP Top 10 vulnerabilities (including complex XSS and SQL injection vectors) in externally exposed client and supplier systems.
- Architected internal security tooling and vendor risk assessment scripts, automating due diligence pipelines for high-risk supply-chain API onboarding.
- Bridged the gap between technical exploits and executive governance, engineering dashboarding systems to visualise security telemetry and enforce NCSC and ISO 27001 control alignment across external infrastructure.

Technical Toolkit

- **Programming & Architecture:** Python (FastAPI), TypeScript (Next.js, Node), Dart (Flutter), Solidity, Shell scripting
- **Data & ML Infrastructure:** XGBoost, Pandas, Memgraph, MongoDB, Redis, Uvicorn
- **Security & CI/CD:** Hardhat, Foundry, Docker, KQL, Zero-Knowledge Proofs (Groth16), API security testing
- **Vulnerability Management Tools:** Nessus, Qualys, Microsoft Defender, bespoke security scripts

Education and Certifications

Bachelor of Technology, Computer Science and Engineering
Ladoke Akintola University of Technology

Selected Qualifications

- CompTIA Security+ (Certified)
- Advanced Software Architecture & Machine Learning (Independent Research & AMTTP Development)

Professional Interests

- Advancing privacy-preserving cryptography (ZK-SNARKs) in enterprise FinTech applications
- Designing low-latency machine learning pipelines for hostile network topologies
- Bridging the technical gap between modern legislative cybersecurity requirements and verifiable software design